MIL-R-45036D(MR)
14 July 1978
SUPERSEDING
MIL-R-45036C(MR)
2 August 1973

MILITARY SPECIFICATION

RUBBER, HARD (EBONITE), NATURAL OR SYNTHETIC, SHEET, STRIP, ROD, TUBING, AND MOLDED PARTS

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for various grades of natural or synthetic hard rubber (ebonite) material as sheet, strip, rod, tubing, molded parts or other forms for use in military equipment (see 6.2).

1.2 Classification. The hard rubber shall be furnished in the following grades as specified (see 6.2).

Grade A - General Purpose
Grade B - Heat Resistant
Grade C - Electrical Resistant
Grade D - Impact Resistant

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.
MIL-STD-297 - Visual Inspection Guide for Hard Rubber (Ebonite) Items.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

FSC 9320, 5970

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D149	- Test for Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D256	- Tests for Impact Resistance of Plastics and Electrical Insulating Materials.
D257	- Test for D-C Resistance or Conductance of Insulating Materials.
D530	- Testing Hard Rubber Products.
D648	- Test for Deflection Temperature of Plastics Under Load.
D2240	- Test for Indentation Hardness of Rubber and Plastics by means of a Durometer.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The material shall be natural or synthetic rubber or a mixture thereof, or recycled or reclaimed rubber, vulcanized with sulfur in quantities sufficient to produce a product having a ratio of combined sulfur to rubber hydrocarbon in excess of 15 percent.

3.2 Physical property requirements. The hard rubber material shall conform to the physical property requirements of table I, II or III, as applicable when tested in accordance with 4.6.

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Table I. Sheet and strip (physical property requirements)

Property	Grade A General Purpose	Grade B Heat Resistant	Grade C Electrical Resistant	Grade D Impact Resistant
Tensile strength, psi, minimum (or)	8000	5000	8000	6000
MPa, minimum	55.2	34.5	55.2	41.4
Elongation, percent, minimum	3	1	3	2
Hardness, Rockwell penetration, maximum	125	115	115	115
recovery, minimum	70	75	70	70
Flexural strength, psi, minimum (or)	14000	-	14000	12000
MPa, minimum	96.5	-	96.5	82.7
Impact resistance ft. lbs./inch of notch, min. (or)	.25	0.20	0.35	.50
Nm/m of notch, minimum	13.3	10.7	18.7	26.7
Dielectric Strength 1/32" (0.08 cm) thick material, 60 cycles Volts/mil, minimum	300	-	400	-
kilovolts/mm, minimum	11.8	-	15.7	-
Cold flow (1,000 lbs. (454 kg.) for 22 hrs) % maximum at 50°C (122°F)	45.0	-	-	1.5
Heat distortion temperature, 264 psi (1.82 MPa) degrees C., minimum (or)	54	121	-	-
degrees F., minimum	130	250	-	-
Insulation resistance after sunlight exposure, megohms, at 35°C ± 1°C (95° ± 2°F) and 90 ± 2% relative humidity, minimum	-	-	20	-
Water absorption, 48 hrs. at Room Temperature percent, maximum	-	-	0.08	-

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Table II. Rod and tube (physical property requirements)

Property	Grade A General Purpose	Grade B Heat Resistant	Grade C Electrical Resistant	Grade D Impact Resistant
Tensile strength,				
psi, min.	6000	6000	5000	3000
MPa, minimum	41.4	41.4	34.5	20.7
Elongation, percent, minimum	3	1	2	3
Hardness, Rockwell				
penetration, maximum	125	-	125	-
recovery, minimum	70	-	70	-
Flexural strength,				
psi, min.	10,000	-	-	-
MPa, minimum	68.94	-	-	-
Impact resistance,				
ft., lbs./inch of notch, min.	.25	0.20	-	.75
(or) Nm/m of notch min.	13.3	10.7	-	40.0
Dielectric strength,				
1/32" (0.08 cm) thick material				
60 cycles				
volts/mil, minimum	300	-	400	-
kilovolts/mm, minimum	11.8	-	15.7	-
Heat distortion temperature,				
264 psi (1.82 MPa)				
degrees C., minimum (or)	54	121	-	-
degrees F., minimum	130	250	-	-
Water absorption,				
48 hours at Room Temperature				
percent, maximum	-	-	0.08	-
Durometer D Range	75-85	75-85	75-85	75-85

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Table III. Molded goods (physical property requirements)

Property	Grade A General Purpose	Grade B Heat Resistant	Grade C Electrical Resistant	Grade D Impact Resistant
Tensile strength, psi, minimum (or)	7000	6000	3000	3000
MPa, minimum	48.2	41.4	20.7	20.7
Elongation, percent, minimum	3	2	1	3
Hardness, Rockwell penetration, maximum	130	110	110	-
recovery, minimum	75	70	55	-
Flexural strength psi, minimum (or)	10,000	-	-	-
MPa, minimum	68.94	-	-	-
Impact resistance, ft. lbs./inch of notch, min.	.25	-	-	.75
(or) Nm/m of notch, min.	13.3	-	-	40.0
Dielectric strength, 1/32" (0.08 cm) thick material, 60 cycles volts/mil, minimum	300	-	400	-
kilovolts/mm, minimum	11.8	-	15.7	-
Heat distortion temperature degrees C., minimum (or)	54	121	-	-
degrees F., minimum	130	250	-	-
Water absorption, 48 hours at Room Temperature percent, max.	-	-	0.08	-
Durometer D Range	-	-	-	75-85

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3.3 Dimensions.- The dimensions of the sheet, strip, rod, tubing and molded parts shall be in accordance with the requirements of the applicable drawings or specifications and the contract or purchase order. Unless otherwise specified, the tolerances shall be within the limits specified in tables IV, V, VI, VII and VIII.

Table IV. Tolerances for molded parts[1/]

Normal dimensions	±0.006 inch per inch
Across parting line	±0.006 inch per inch, plus 0.020 inch

1/ Unless otherwise specified on the part drawing.

Table V. Tolerances for extruded rod and tubing (ground or polished)

Outside diameter, inch	Tolerance, inch Grades A, B and C
Up to 3/8 incl	+0.001 to -0.001
Over 3/8 to 3/4 incl	+0.002 to -0.001
Over 3/4 to 1 incl	+0.002 to -0.002
Over 1 to 1-1/4 incl	+0.003 to -0.002
Over 1-1/4 to 1-1/2 incl	+0.003 to -0.003
Over 1-1/2 to 2 incl	+0.004 to -0.004
Over 2 to 3 incl	+0.005 to -0.005
Over 3 to 3-1/2 incl	+0.006 to -0.006
Over 3-1/2 to 4 incl	+0.007 to -0.007
Over 4 to 4-1/2 incl	+0.008 to -0.008
Over 4-1/2 to 5 incl	+0.015 to -0.015

Table VI. Tolerances for extruded rod and tubing (rough)

Outside diameter, inch	Tolerance, inch	
	Grade A	Grades B and C
Up to 1/8 incl	+0.015 to +0.029	+0.013 to +0.031
Over 1/8 to 1/4 incl	+0.020 to +0.033	+0.018 to +0.035
Over 1/4 to 3/8 incl	+0.025 to +0.036	+0.023 to +0.038
Over 3/8 to 1/2 incl	+0.025 to +0.041	+0.023 to +0.044
Over 1/2 to 5/8 incl	+0.025 to +0.044	+0.022 to +0.047
Over 5/8 to 3/4 incl	+0.025 to +0.048	+0.022 to +0.052
Over 3/4 to 1 incl	+0.025 to +0.055	+0.021 to +0.060
Over 1 to 1-1/4 incl	+0.025 to +0.062	+0.020 to +0.068
Over 1-1/4 to 1-1/2 incl	+0.030 to +0.070	+0.024 to +0.076
Over 1-1/2 to 1-3/4 incl	+0.030 to +0.077	+0.023 to +0.084
Over 1-3/4 to 2 incl	+0.035 to +0.085	+0.028 to +0.093
Over 2 to 2-1/2 incl	+0.040 to +0.090	+0.033 to +0.098
Over 2-1/2 to 3 incl	+0.045 to +0.095	+0.038 to +0.103
Over 3 to 3-1/2 incl	+0.050 to +0.100	+0.043 to +0.108
Over 3-1/2 to 4-1/4 incl	+0.060 to +0.125	+0.050 to +0.135
Over 4-1/4 to 5 incl	+0.073 to +0.135	+0.063 to +0.145

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Table VII. Tolerances for sheet and strip

Thickness inch	Tolerance, inch			
	Grade A	Grade B	Grade C	Grade D
0.005	-0.000 to +.002	1/	2/	2/
Over 0.005 to 0.010 incl	-.000 to +.003	1/	-.000 to +.004	±.003
Over 0.010 to 1/64 incl	-.000 to +.004	1/	-.000 to +.005	±.003
Over 1/64 to 1/32 incl	-.000 to +.005	-.000 to +.005	-.000 to +.008	±.004
Over 1/32 to 1/16 incl	-.000 to +.008	-.000 to +.009	-.000 to +.015	±.006
Over 1/16 to 1/8 incl	-.000 to +.011	-.000 to +.016	-.000 to +.018	±.010
Over 1/8 to 1/4 incl	-.000 to +.014	-.000 to +.027	-.000 to +.024	±.016
Over 1/4 to 3/8 incl	-.000 to +.016	-.000 to +.034	-.000 to +.030	±.020
Over 3/8 to 1/2 incl	-.000 to +.019	-.000 to +.040	-.000 to +.035	±.025
Over 1/2 to 5/8 incl	-.000 to +.021	-.000 to +.044	-.000 to +.040	±.030
Over 5/8 to 3/4 incl	-.000 to +.026	-.000 to +.045	-.000 to +.045	±.035
Over 3/4 to 7/8 incl	-.000 to +.030	-.000 to +.045	-.000 to +.045	±.045
Over 7/8 to 1 incl	-.000 to +.032	-.000 to +.050	-.000 to +.055	±.045

1/Not supplied under 1/32 inch.

2/Not supplied under 0.010 inch.

Table VIII. Tolerances for extruded tubing (rough ground or polished)

Inside diameter, inch	Tolerance, inch	
	Grade A	Grades B, C and D
Up to 0.125 incl	+0.005	±0.003
Over 0.126 to 0.500 incl	+0.006	±0.004
Over 0.501 to 0.750 incl	+0.009	±0.006
Over 0.751 to 1.000 incl	+0.012	±0.010
Over 1.001 to 1.25 incl	+0.015	±0.014
Over 1.26 to 1.50 incl	+0.018	±0.016
Over 1.51 to 2.00 incl	+0.024	±0.020

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3.4 Color. Unless otherwise specified, the color of the hard rubber material for Grades A, C and D shall be black and for Grade B either brown or black (see 6.2).

3.5 Finish. The finish of the sheet, strip, rod, tubing and molded parts shall be as specified in the contract or purchase order (see 6.2).

3.6 Identification marking. When specified in the contract or purchase order, identification marking shall be molded on molded parts as specified by the procuring activity (see 6.2).

3.7 Workmanship. Sheet, strip, rod, tubing and molded parts shall be free from dirt, grit, and metal particles as good commercial practice shall permit. Surfaces shall be smooth, free from metal glass, cracks or other manufacturing imperfections which affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all material of one grade, form and size, manufactured from the same raw materials and of the same production run and offered for acceptance at one time.

4.2.1 Unit of product. The unit of product shall be one sheet, strip, rod, tube or molded part.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of units of product shall be selected, from each lot in accordance with MIL-STD-105. Unless otherwise specified inspection level II (normal) shall be used.

4.3.2 Sampling for test. Samples for testing shall be selected at random from each lot. The number of units of product selected shall be sufficient to permit the performance of all the tests specified in 4.6.

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4.4 Examination of units of product. Each of the units of product selected in accordance with 4.3.1 shall be examined for the defects and by the method shown in table IX, to verify compliance with the requirements of this specification. Standards MIL-STD-177 and MIL-STD-297 shall apply as guidance in determining defects. In terms of percent defective the acceptable quality level shall be 1 percent major defects and 2.5 percent total defects (major and minor combined).

Table IX. Classification of defects

Categories	Defects	Requirement paragraph	Method of inspection
Major			
101	Improper dimensions	3.3	Measure with scale graduated to 0.001 in.
102	Embedded dirt, grit or metallic foreign material	3.7	Visual
103	Large blisters	3.7 & 4.4	Visual
104	Large cracks	3.7 & 4.4	Visual
Minor			
201	Improper color	3.4	Visual
202	Improper finish	3.5	Visual
203	Improper identification marking	3.6	Visual

4.5 Inspection of packaging. Preservation, packaging, packing and marking for shipment shall be examined for compliance with section 5 requirements.

4.6 Tests

4.6.1 Test specimens. Test specimens for all tests shall be prepared from sample units selected in accordance with 4.3.2 and in accordance with the methods shown in table X and 4.6.1.1 and 4.6.1.2.

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Table X. Test specimens and test procedures

Property	ASTM Method
Tensile strength	D530
Elongation	D530
Hardness, Rockwell	D530
Flexural strength	D530
Impact resistance	D530 and D256
Dielectric strength	D149
Cold flow	D530
Heat distortion temperature	D648
Durometer hardness	D2240

4.6.1.1 Specimens for insulation resistance after sunlight exposure. The test specimen shall be in sheet form not less than 3/16 inch in thickness by 2-1/2 inches wide by 6 inches long. It shall be drilled for electrodes in accordance with figure 1 of ASTM D257. The specimen shall be sandpapered on both sides by hand with No. 1 sandpaper to remove the molded surface and washed in petroleum ether, scrubbed in hot water with a stiff bristled brush, rinsed with distilled or deionized water and allowed to dry at room temperature in dark or dim light. Brass parts shall be washed in petroleum ether, then attached to the specimen as shown in figure 1 of ASTM D257.

4.6.1.2 Specimens for water absorption. Two test specimens shall be ground to approximately 2 inches by 2 inches by 1/4 inch thick. The specimens shall be washed in running water and dust particles removed with a wet brush.

4.6.2 Test procedures. Test specimens prepared in accordance with 4.6.1 shall be tested in accordance with the methods shown in table X, 4.6.2.1 and 4.6.2.2 in order to determine conformance with the requirements of 3.2.

4.6.2.1 Insulation resistance after sunlight exposure.

4.6.2.1.1 Apparatus. The light source shall consist of a sun lamp mounted in an Admedium socket, energized through a transformer with stabilized 100 volts. The lamp shall be aged by normal operation for a period not less than 250 hours and shall be discarded after a life of 2000 hours.

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4.6.2.1.2 Light exposure. The specimen (see 4.6.1.1) shall be suitably supported parallel to the lamp axis at a distance of 12 inches with the faces in normal position, the centers in the same horizontal plane, care being taken that the specimen falls within the shadow of the supporting wires within the lamp bulb. Another preferred arrangement is to support the specimens on the periphery of a circular cage concentrically around the lamp. With specimen in place, expose to light for 200 hours at room temperature.

4.6.2.1.3 Resistance between electrode pairs. The test specimen with the electrodes shall be conditioned for not less than 96 hours at 35° ± 1.1°C (95° ± 2°F) and 90 ± 2 percent relative humidity. The insulation resistance between pairs of electrodes shall be determined in accordance with ASTM D257, using 100 volts d.c.

4.6.2.2 Water absorption test procedure. Specimens prepared as specified in 4.6.1.2 shall be rinsed in distilled water, dried at 100° ± 1.1°C (212° ± 2°F) for one hour, cooled to room temperature in a desiccator and weighed. Each specimen shall be placed in a separate beaker of distilled water for 48 hours at room temperature. A small piece of glass rod or tubing shall be placed in the beaker underneath the specimen to provide space for free access of water to the lower surface of the specimen. At the end of the 48 hours, the specimens shall be removed from the water, wiped dry and weighed immediately. Duplicate tests shall be run and the average of the two tests shall be used for determining compliance with the requirement of tables I, II or III as applicable. The percent water absorbed is determined as follows:

$$\frac{\text{Final weight - original weight}}{\text{original weight}} \times 100$$

4.7 Rejection.

4.7.1 Examination. Any nonconforming sample unit in each sample when examined in accordance with 4.4 shall be rejected and if the nonconforming units exceed the acceptance number for that sample, the entire lot shall be rejected.

4.7.2 Tests. A lot shall be rejected for failure to comply with any of the test requirements when tested in accordance with 4.6.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Commercial. Hard rubber material shall be preserved and packaged in accordance with MIL-STD-1188.

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5.2 Packing.

5.2.1 Commercial. Hard rubber material packaged as specified in 5.1.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking for shipment. In addition to any special marking required by the contract or order, all shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 General purpose grade. The general purpose grade is intended for use where mechanical and minimum electrical resistance is required.

6.1.2 Heat resistant grade. The heat resistant grade is intended for use up to 200°F as compared to 160°F for the general purpose grade.

6.1.3 Electrical resistant grade. The electrical resistant grade is intended for the maximum in electrical resistance.

6.1.4 Impact resistant grade. The impact resistant grade is intended for use where shock loading is encountered.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Form of material required (see 1.1).
(c) Grade of material required (see 1.2).
(d) Dimensions required (see 3.3).
(e) Color if other than black for grades A, C and D and other than brown or black for grade B (see 3.4).
(f) Finish (see 3.5).
(g) Identification marking (see 3.6).
(h) Special marking (if required) (see 5.3).

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6.3 Metric units. When metric dimensions are required, units for inch and foot may converted to the metric equivalent by multiplying them by the following conversion factors:

English unit	Multiply by	Equals	Metric SI equivalent
Inch	0.0254	=	Metre (M)
Foot	0.3048	=	Metre (M)

NOTE: Conversion factors can be associated with ASTM E 380 entitled "Metric Practice Guide".

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-R-45036D(MR), Rubber, Hard (Ebonite), Natural or Synthetic, Sheet, Strip, Rod, Tubing, and Molded Parts

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